

Football Players' Market Value Prediction

Giacomo Cordella

Bocconi University

Abstract—The scope of the project was to accurately approximate the market value of real football players given their statistics and more information about them.

To do this, a dataset with features concerning all aspects of a player's characteristics was given to be analyzed and used, in order to train a Machine Learning algorithm capable of predicting their market value in euros. To achieve this goal, the approach taken can be subdivided into smaller tasks, which tackle the main objective.

The data set was first analyzed in its entirety, looking at the variables carrying the most information about the value of a player, trying to understand if the variable was totally useless, it needed to be manipulated (engineered) in order to be actually informing, or was already telling much about the player's value. This was the most important part that concerned Feature Engineering and Selection and Data Processing, used to fill the gaps left by missing data.

The second part concerned Model Selection: here many models, both supervised and unsupervised, were trained and tested onto respectively train and validation set, to understand their efficiency and their power when approximating the players' market values. Subsequently, the best model underwent to a cross validation to ensure its consistency and efficacy on the whole training set.

Lastly the model selected was retrained on the whole, already optimized and engineered training set and used to predict the values of never seen players from the given test set.

The results obtained from the evaluation of the best model were astonishing, meaning that the choice of using that model was correct and that, most probably, with a relatively low error margin, the prediction made should be correct.

I. DATASET DESCRIPTION

The data given for the project were composed by a *training* and *test* datasets, a *columns description* file which described all the features present in the datasets and a *submission* example that had to be replicated when submitting the results.

The *train* set contained 15391 data points with 75 features each, while the *test* set contained 3848 data pointing with 74 features with the only one missing being the market value as expected.

These features, described in the *columns description* file, could be firstly subdivided in numerical and categorical features, then could be subdivided into other categories as follows:

- **Person related information:** *name, height, weight, age, date of birth, ...*

- **Player related information:** *overall, potential, role played, club playing for, role played in the club, international reputation, ...*

- **Playing related information:** *playing traits and tags, all playing characteristics, work rate, ...*

For what concerns the scope of the project some of these variables are useless for our model, other might be useful but need some refinement in order to convey actual information for our model, therefore the following step would be to understand which variables should be kept or dropped.

II. FEATURE SELECTION

The Feature Selection step is the most important step of the project, here not only information are chosen to be given to a model, but also variables are analyzed and engineered in order to adapt them to the context, trying to take the most information possible out of them and cope with the "course of dimensionality" which leads to sparseness in our sample space.

This fundamental step can be subdivided into smaller steps that helped to achieve a good feature selection:

- **First screening** of the totally useless features and their elimination
- **Data engineering** of those features that contain indirect information
- **Elimination** of those data points that have no values for the players' market value
- **Imputation** for those data points that have missing values in other features
- **Analysis** of the features and their correlation with the target feature
- **Final feature selection**

After these steps, the output should be a polished, low dimension, and essential dataset ready for the training and evaluation of the models.

A. Data Screening

In this initial step the useless features are dropped because not containing any or vary low information about the market value of a player. Indeed, the features dropped in this step are the followings: *id*, *short name*, *long name*, *dob*, *club name*, *league name*, *club jersey number*, *club joined*, *nationality name*, *nation jersey number*, *club team id*, *nationality id*, *preferred foot*, *work rate*, *body type*, *player tags*, *player traits*.

Even if some of these variables may be taken into consideration for evaluating the market values of a player, as for example the club name, work rate or preferred foot, the influence that these variables would have on a player's value would not be much relevant if not in really extreme and rare cases, in which, either ways, other variable would be much more explicative than these.

B. Data Engineering

Taking a look to the dataset, an important part of the features included in it are categorical, e.g. *player position*, *nation position*, *club position*, etc. After the first screening, the remaining categorical variables that carry indirect information (or that need some manipulation to be actually informative) are: *player position*, *nation position*, *club loaned from*, *club position*, *contract valid until*.

The reasoning behind this selection is the following:

- **Player Position**, the role of a player can tell something about his value, e.g. a good striker can value much more than a good defender;
- **Nation Position**, a player who plays in the National team usually is a very good player and can value more than a player who does not play for the National team;
- **Club Loaned From**, when a player is on loan means that is not good enough for his main team but better than average in the team is playing in and this information can be important for evaluating the value of a player;
- **Club Position**, a player who actually plays in a team is more valuable than a player who is a substitute or even a reserve;
- **Contract Valid Until**, a player with an expiring contract usually tend to cost less then someone as good as him but with a just signed contract.

These features are then modified to take out the information just listed rather the one they are already conveying, and then added to the dataset, eliminating the old columns. Now that the dataset has all the useful features need to undergo an analysis to understand which are the best features for our task.

C. Elimination and Imputation

The first analysis made consist in finding how many player did not have a market values, because the players with no market value were just 58 (very few compared to the total amount of players in the dataset) these data points were dropped. The other *NaN* values where substituted with a 0, made exception for the *release clause* that, being an important variable for evaluating the market value of a player, was imputed with the mean of the other data points.

Having polished also the dataset, the real analysis could take place.

D. Features Analysis

The analysis made on the dataset was a **Correlation Analysis**. This was made in order to understand the correlation relation between features and , in particular, between the market value and the other features. Especially in the second case, this analysis was useful to see which features were more correlated to the target feature, meaning that the most correlated variables were good linear predictors for player's values.

NOTE: the correlation coefficient only checks for linear correlation between variables and does not detect other non-linear correlations, leaving out all those non-linearly correlated variables. However, despite being the simplest measure of correlation, it showed to be efficient, simple to compute and a very useful for predictions. Therefore, the analysis might lack of polynomial analysis that maybe would have brought more accurate predictions but the computational cost associated to the increase in correctness would have been unjustified, so, for Ockham's razor, the efficient and effective simplicity was chosen.

After having computed the correlation matrix for the whole dataset and all features, the results shows:

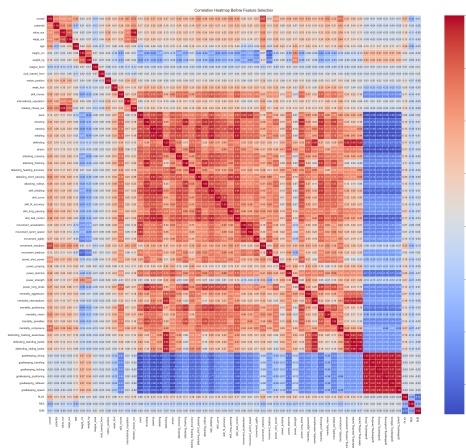


Fig. 1: Heatmap Before Processing

From this heatmap not much information can be extracted, therefore the next step of the analysis was to take the most correlated features with target feature, i.e. market value.

The metric used to build the other correlation matrix was to find all those features that satisfied this relation:

$$\left| \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sigma_x \sigma_y} \right| \geq 0.25$$

This means that the variables desired must have a correlation coefficient greater than 0.25 in absolute value, checking both positive correlation and negative correlation.

In addition to this, those variables which are highly correlated with each other are going to be discarded, in order to avoid multicollinearity and repetition of information, i.e. all those variables with a correlation coefficient with each other over 0.90 are going to be removed.

E. Final Selection

Therefore, for the final feature selection, the variables chosen are the one present in the after processing heatmap obtained as just said in the previous section:

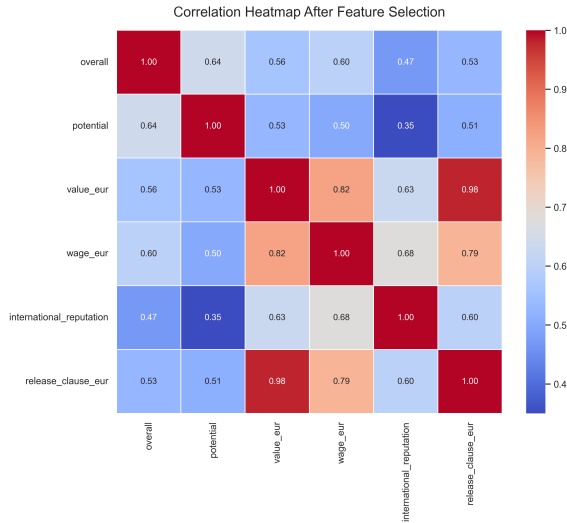


Fig. 2: Heatmap After Processing

From this heatmap we can see that the most important features are just 5: **Overall, Potential, Wage, International Reputation and Release Clause**.

Taking only these 5 variables and dropping the others, the dimension of the dataset have been reduced significantly, reducing also the sparseness of the data points, making the work easier for the model to train.

Therefore, the final dataset has been computed and is now ready for the training stage.

III. MODEL SELECTION AND ENGINEERING

Having found the best features, they can be applied to both the train and test sets. From here the model engineering and model selection can start, and the models chosen to be trained and evaluated are the following:

- **Linear Regression**, a simple linear regression;
- **Ridge Regression**, a Linear Regression with L2-Regularizer;
- **Lasso Regression**, a Linear Regression with L1-Regularizer;
- **Elastic Net Regression**, a Linear Regression with a mix of L2 and L1 Regularizers;
- **AdaBoost**, a boosting ensemble methods made on Simple Decision Trees;
- **Gradient Boosting**, a different boosting ensemble method;
- **Random Forest**, a bagging ensemble learning method that operates by constructing multiple decision trees during training and outputs the average prediction.;
- **Simple Neural Network**, a simple neural network with just one hidden layer, fully connected and with ReLu activation, and a patience for early stopping.

To train the models, the training set was split into two subsets, a validation set, containing 20% of the total data points taken at random, and a train set containing the remaining 80% of data points from the initial train set.

The comparison between models has been made after manually tuning the hyper-parameters of supervised and unsupervised machine learning models (more efficient way than running Grid Search or Random Search for each model)

It's important to note that, for maximum accuracy, each model should ideally be tuned using a hyperparameter optimization method such as Grid Search or Random Search. This process would involve testing different combinations of hyper-parameters and evaluating each model's performance using k-fold cross-validation to ensure robustness and generalizability. However, such an approach is extremely computationally intensive and time consuming, it could take over a week to complete even a single full run.

Given these constraints, a more efficient and practical strategy was adopted: manual tuning of the most relevant hyperparameters. This allowed for a good balance between model performance and computational feasibility.

IV. MODEL EVALUATION

For the models evaluation the training set was split into train and validation sets, as already mentioned and the error was calculated through the **Root Mean Squared Error** metric. According to this metric, the best performing model (lowest **RMSE**) was chosen as the best model, and passed to the second step.

From this first step of the evaluation, the best Model was the **Simple Neural Network**, from which the decrease in RMSE through the epochs was recorded and showed in a graph

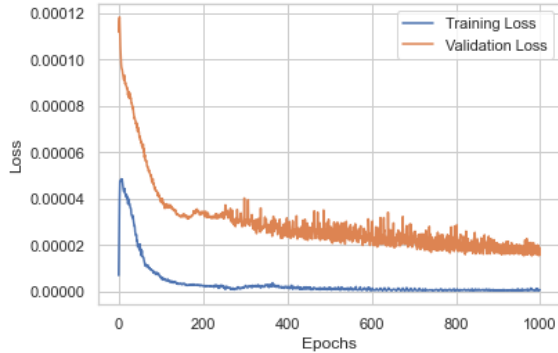


Fig. 3: SNN Learning Curve

The best model found, then, underwent a Cross Validation in which the training set was split into k folds and, for each fold, a fold constituted the validation set, onto which validating the model, and the remaining folds the train set, onto which training it. A record of the Root Mean Squared Error made by the model for each fold was kept, and, at the end of the folds rotation, the error mean was calculated, showing the consistency of the model chosen.

In this phase, the Simple Neural Network averaged a RMSE of around 350000€ among 5 folds, which is a pretty amazing result taken into consideration that many players' market values are in the order of millions of euros.

V. FINAL RESULTS

The final step is now actually predicting the players' market values in the test set given at the beginning of the project and save the results in a submission-like file.

For this last step the test set need to undergo the same data processing and feature selection of the train set, this will allow the model to correctly predict the values.

Another important step is to retrain the best model, the Simple Neural Network, not anymore on a test subset as done up to this moment, but on the whole training set given at the beginning of the project. In this way, the whole dataset is exploited to better even more the model, that now can be used to predict the market values of the players in the test set and save these predictions on in a CSV File that will be submitted.

VI. CONCLUSION

This project successfully addressed the challenge of predicting the market value of professional football players using machine learning techniques. The entire pipeline, from data cleaning and transformation to model training and evaluation, was carefully designed to maximize for what possible the predictive accuracy while maintaining computational efficiency.

The initial dataset analysis focused on identifying and removing non-informative features. This step was crucial in reducing dimensionality and eliminating noise, especially given the presence of many categorical and identifier-type variables. Through feature engineering, several categorical features (e.g., player position, national team presence, loan status, and contract details) were transformed into meaningful numerical indicators. A further correlation analysis identified the features most linearly related to the market value, ultimately narrowing down to a core set of highly informative predictors: Overall, Potential, Wage, International Reputation, and Release Clause.

In the model selection phase, multiple regression models were explored—ranging from linear regressions with regularization (Ridge, Lasso, Elastic Net) to ensemble methods (AdaBoost, Gradient Boosting, Random Forest) and finally to a Simple Neural Network. The neural network, featuring a single hidden layer and early stopping, consistently outperformed other models based on the Root Mean Squared Error (RMSE) metric.

After identifying the best-performing model, a cross-validation strategy was employed to confirm its robustness, with the neural network achieving an average RMSE of approximately €350,000. This level of error is relatively small when compared to the multi-million euro scale of many player valuations, underscoring the model's practical effectiveness.

For final deployment, the Simple Neural Network was retrained on the full training set and used to predict values on the test set, with results saved in the required submission format.

In conclusion, this project demonstrates a successful application of machine learning to a real world problem, with well-reasoned modeling choices and a strong focus on data quality, interpretability, and generalization.

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